INCH-POUND
MIL-R-83726/24K(USAF)
23 June 1995
SUPERSEDING
MIL-R-83726/24J(USAF)
9 August 1991

MILITARY SPECIFICATION SHEET

RELAYS, SOLID STATE, TIME DELAY (ON OPERATE), TRACK MOUNTED, TYPE I, CLASS C, SPST, 150 MILLIAMPERES, FIXED TIME, 50 MILLISECONDS TO 500 SECONDS

(K) Inactive for new design after 23 June 1995. No superseding document.

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-83726.

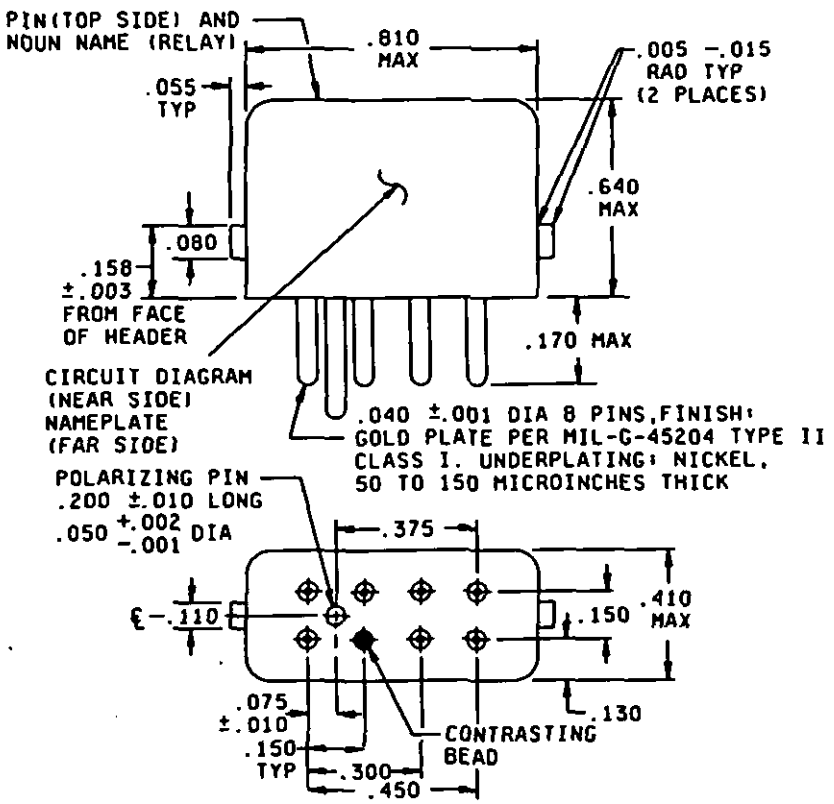
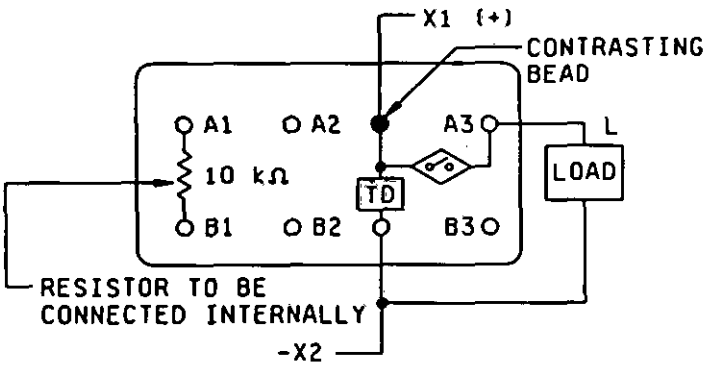


FIGURE 1. Dimensions and configuration.

(K) denotes changes

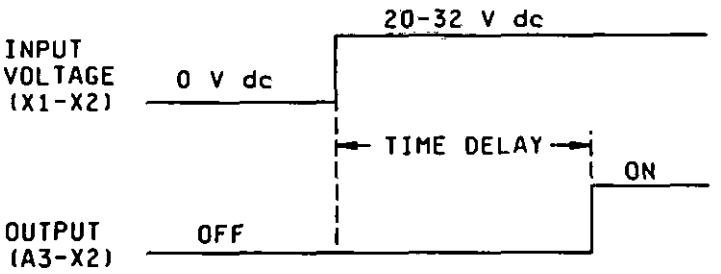
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CIRCUIT DIAGRAM

(SEE NOTES 4 AND 5)



TIMING DIAGRAM

Inches	mm	Inches	mm
.001	0.03	.130	3.30
.002	0.05	.150	3.81
.003	0.08	.158	4.01
.005	0.13	.170	4.32
.010	0.25	.200	5.08
.015	0.38	.300	7.62
.040	1.02	.375	9.52
.050	1.27	.410	10.41
.055	1.40	.450	11.43
.080	2.03	.640	16.26
.110	2.79	.810	20.57

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
3. Metric equivalents are given for general information only.
4. Terminals A1, A2, B1, and B2 are connected internally. Do not use for external tie points or for terminals.
5. There shall be affixed to the relay a legible circuit diagram that identifies each terminal location specified.
6. Relays are intended for use with socket mount MIL-R-6106/23-001 and bracket mount MIL-R-6106/25.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS:

Operating data:

Timing action: Delay-on operate.

Time delay: Fixed; select from .05 to 500 seconds.

Timing accuracy: ±10 percent of the nominal value. 1/

Recycle time: 10 milliseconds. 2/

Power interrupt: The accuracy will not be affected by power interruptions up to 1 millisecond spaced at least 10 milliseconds apart.

Input data:

Input voltage: 28 V dc nominal; range 20 V dc to 32 V dc.

Duty rating: Continuous.

Current drain: 10 milliamperes maximum plus load at +25°C.

Polarity protection: The timer shall be inoperative during, and undamaged by, reversal of the polarity of the operating voltage.

Output data:

Configuration: SPST; switch closure to +28 V dc.

Rating: 150 milliamperes maximum.

Suppression: Inductive suppression provided for output protection.

Voltage drop: 2 V dc maximum.

Leakage current: 1.0 microampere maximum at 28 V dc and +25°C; 10 microamperes maximum at 28 V dc and +125°C.

Endurance (life): 1,000,000 cycles minimum for test.

Qualification: 2,000 hours.

Electrical data:

Transients: 3/

Positive transients: +80 volts in accordance with MIL-STD-704, figure 9, limit 1.

Self-generated transients: ±50 volts maximum.

Spike susceptibility: ±600 volts, 10 microseconds maximum.

1/ The accuracy requirement applies to any combination of operating temperature and voltage.
2/ Recycle time is defined as the minimum time that power must be removed from the input terminals to assure that the next timing cycle will be completed within the specified timing tolerance. (Units can be recycled during timing or after time-out.)
3/ Transient specifications are based on a maximum duty cycle of 1/50.

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Electromagnetic interference: In accordance with MIL-STD-461, class 1D. 4/

Insulation resistance: 1,000 megohms at 500 V dc between each pin and case. 5/

Dielectric strength: 1,000 volts rms at 60 Hz at sea level between case and pins connected together. 5/

Environmental data:

(K) Maximum altitude rating: 80,000 feet.

Ambient temperature (operating and nonoperating): -55°C to +125°C.

Vibration (sinusoidal): 10 Hz to 55 Hz at 0.06 inch DA; 10 g's, 55 Hz to 500 Hz; 20 g's, 500 Hz to 3,000 Hz.

Vibration (random): MIL-STD-202, method 214, condition I, letter C, duration of 30 minutes.

Shock: 1,100 g's for 0.5 millisecond, 1/2 sine, 3 axes.

Acceleration: 100 g's, 3 axes.

Seal: Hermetic.

Moisture resistance: MIL-STD-202, method 106.

Physical data:

Dimensions and configuration: See figure 1.

Terminations: See figure 1.

Terminal strength: 3 pounds pull.

(K) Weight: 0.56 ounce (15.7 grams) maximum.

Marking: See MIL-R-83726. In addition, relays shall be marked with the ESDS identifier as specified in MIL-STD-1285.

ESDS protection program: An ESDS protection program shall be implemented within 6 months of the date of revision H to this document. The manufacturer shall establish and maintain an ESD control program in accordance with MIL-STD-1686 for mission critical equipment. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification. This program shall be documented by an ESD control plan which must be under document control. As a minimum, this plan must address the identification of ESDS sub-components and end items, facilities, training, design protection, handling procedures, marking, cleaning, preservation, packaging, and quality assurance. A model ESD control program is available from the qualifying activity and may be used as a guideline. Further guidance for ESD control is available from the EOS/ESD Association and the Electronics Industry Association (EIA). This requirement is applicable to all manufacturers who handle ESDS component parts and materials in the relay manufacturing or testing process. This requirement is not limited to manufacturers qualifying ESDS end items.

4/ EMI test limits shall not be exceeded during the timing interval or when continuously energized under steady-state conditions in accordance with the radio interference requirements of MIL-R-83726.

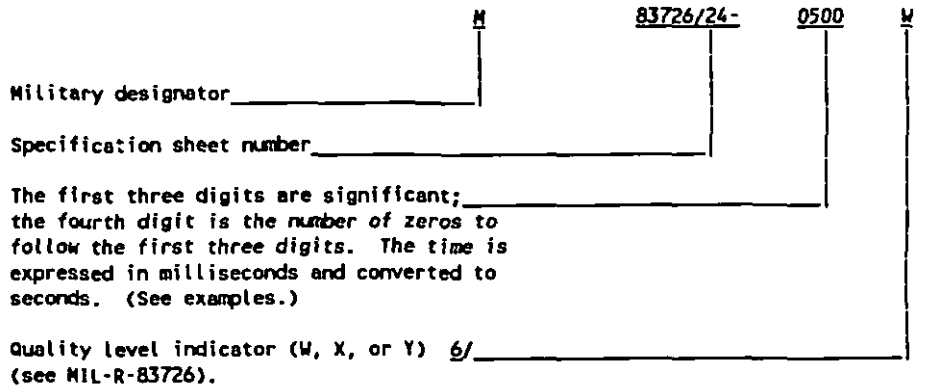
5/ Terminals X1, X2, and A1, A3, and B1 must be connected together during this test. Dielectric withstanding voltage and insulation resistance are measured between all mutually insulated terminals and the case.

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ESDS verification: As a part of qualification or qualification after redesign, ESD testing shall be done in accordance with method 3015 of MIL-STD-883 modified to test at 16,000 volts. Testing at lower voltage levels is not required. This testing shall be accomplished as part of the group III for qualification inspection and as part of the group C inspection.

ESDS preservation and packaging: Relays shall be preserved and packaged in such a manner as to ensure that the integrity of ESD sensitive relays is not diminished. ESD sensitive relays shall be preserved and packaged in accordance with the requirements of MIL-STD-1686.

Part or Identifying Number (PIN): Consists of the prefix M83726/24-, a four-digit dash number (expressed in milliseconds), and a quality level indicator as follows:



Examples: M83726/24-0500W - 50 millisecond time delay W level.
M83726/24-2503X - 250 second time delay X level.
M83726/24-5003Y - 500 second time delay Y level.

(K) Quality assurance provisions: Group B and group C testing are not required. The manufacturer shall notify the qualifying activity in the event of any design or construction changes, and shall impose additional testing requirements as necessary.

(K) The Qualified Products List (QPL) associated with this inactive for new design specification will be maintained until acquisition of the product is no longer required, whereupon the specification and the QPL will be canceled.

6/ Any relays numbered prior to the date of this specification without a quality level indicator shall be considered interchangeable (store and issue) with the "W" quality level.

CONCLUDING MATERIAL

Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-F775-02)

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